KendraFi

Project Overview & System Architecture

Weiyedai-Global • Master Context & Agent Guide

Competition	2026 Shenzhen International FinTechathon — International Track
Theme	AI as a Financial Participant
Focus	Topic E: SME Finance Copilot • Topic F: Agent Wallet bonus
Submission deadline	October 20, 2026, 23:59 UTC+8

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Competition: 2026 Shenzhen International FinTechathon (International Track)[cite: 1, 2]

Co-Hosts: Tsinghua University PBC School of Finance (PBCSF) & WeBank[cite: 1, 2]

Official Theme: "AI as a Financial Participant" — Topic E (SME Finance Copilot) + Topic F (Agent Wallet +5 Bonus)[cite: 1, 2]

Submission Deadline: October 20, 2026, 23:59 UTC+8 (21:59 Dhaka Time, UTC+6)[cite: 1, 2]

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1. Executive Project Overview

1.1 The Core Thesis in One Sentence

KendraFi (Weiyedai-Global) is an autonomous multi-agent cooperative risk, privacy-preserving underwriting, and programmable B2B escrow protocol that bridges Grameen Bank’s Nobel-winning 5-person peer syndicate (Kendra) economics with WeBank’s AI-Native ABCD (AI, Blockchain, Cloud, Data) architecture[cite: 1, 2].

1.2 Dual Problem Statement (Dhaka Street Markets vs. Shenzhen Digital Banking Export)

Perspective A: The Dhaka / Emerging-Market Reality (The Borrower's Trap)

Imagine Rahim, who runs a profitable retail grocery stall in Karwan Bazar, Dhaka, Bangladesh, and needs $400 (45,000 BDT) every two weeks to restock 50 sacks of Miniket rice and cooking oil from his wholesale distributor[cite: 1, 2]. Despite five years of zero defaults, Rahim is trapped between two broken options[cite: 1, 2]:

1. Traditional Physical Microfinance NGOs (Grameen Bank, BRAC, UDPS) Charge 24%–27% APR: Grameen Bank proved that organizing 5 neighboring shopkeepers into a peer group (Kendra) where members vouch for one another achieves >96% repayment without physical collateral[cite: 1, 2]. However, Bangladesh's 693 MRA-licensed MFIs (disbursing BDT 2,493.02 billion annually) still rely on 1980s human labor: human field officers riding motorcycles to weekly physical meetings, inspecting paper passbooks, and collecting cash[cite: 1, 2]. Paying for thousands of field officers consumes 12%–15% of the loan book, forcing honest merchants to pay 24%–27% effective APR (compared to 9%–16% in formal banking) just to subsidize human bureaucracy[cite: 1, 2].

2. Solo Digital Nano-Loan Apps Fail on Moral Hazard & Cash-Out Caps: When FinTech apps lend to solo individuals on smartphones without the 5-person peer group, they lose Ghatak (1999) assortative peer screening and Stiglitz (1990) peer monitoring[cite: 1, 2]. Unscreened solo borrowers frequently divert cash into personal consumption or "debt-roll" across apps, driving Non-Performing Loans (NPLs) to 10%–18%[cite: 1, 2]. Worse still, Bangladesh enforces a $119/day (10,000 BDT) Mobile Financial Services (MFS) cash-out limit and charges a ~1.85% withdrawal fee—forcing a merchant borrowing 50,000–100,000 BDT into personal bKash/Nagad wallets to visit agents up to ten separate days just to withdraw cash to pay a wholesaler[cite: 1, 2].

Perspective B: The China & WeBank Reality (The "Cold-Start Data Wall")

In Shenzhen, WeBank serves 424 million retail customers and 1.6 million+ MSMEs at an annual IT operation and maintenance (O&M) cost of RMB 2 (~$0.30 USD) per account, 99.99975% availability, and a 1.46% overall NPL ratio (~1.0% on microloans)[cite: 1, 2]:

1. Domestic Weiyedai Mechanics: Inside Mainland China, WeBank's flagship SME product (Weiyedai) and sole-proprietor product (Weihudai) approve collateral-free working capital loans in minutes with a 62% approval rate[cite: 1, 2]. They achieve this by combining FATE Federated Learning (IEEE P3652.1 Heterogeneous Logistic Regression trained on 7.7 billion enterprise entries in SME Large Model 2.0) with official digital VAT tax invoices (X_B) and People's Bank of China (PBOC) credit bureau records (Y), backed by a 20% WeBank capital / 80% partner-bank joint-lending structure and RMB 77 billion in online bulk guarantee services[cite: 1, 2].

2. The Belt-and-Road Bottleneck: In June 2024, WeBank established its international technology subsidiary in Hong Kong—WeBank Technology Services (WeTechnology Limited), capitalized with 150 million USD—to export its digital banking stack along the Belt and Road Initiative (e.g., partnering with SCBX and KakaoBank for Thailand's virtual bank license)[cite: 1, 2]. However, in cash-heavy Belt-and-Road markets (Bangladesh, Indonesia, Thailand, Pakistan, Kenya), where Bangladeshi MSMEs alone face a 2.8 billion financing gap (IFC), 85%–90% of nano-merchants operate outside the formal tax

net[cite: 1, 2]. Because they have zero digital VAT tax filings (X_B) and zero central bank credit bureau labels (Y), Weiyedai hits a "Cold-Start Data Wall"[cite: 1, 2].

1.3 Six Core Innovations KendraFi Introduces

1. 5 Autonomous AI Agents Acting as True Financial Participants: Replaces the human field officer with 5 personalized SME Copilot Agents (‘AGT-01’ to ‘AGT-05’) that audit local merchant cash flows and negotiate fractional joint-liability stakes (\alpha_j \le 0.40) via a 3-Round Contract Net Protocol (CNP) grounded in Besley-Coate (1995) game theory[cite: 1, 2].

2. Belt-and-Road Data Adapter for WeBank FATE & 20/80 Joint Lending: Converts crumpled wholesale paper receipts, Bangla voice notes, and bKash/Nagad MFS SMS logs into clean FATE Hetero-LR feature vectors (X_A: wholesale inventory turnover velocity; X_B: net MFS cash-flow stability) and synthetic peer-stake credit labels (Y), enabling WeBank's 20%/80% joint-lending model to achieve a +12% AUC lift, <1.8% NPLs, and a 12.50% effective borrower APR[cite: 1, 2].

3. Zero-Knowledge Neighbor Privacy (‘secp256k1’ Pedersen + 16-Bit Range Proof) & Supabase RLS: Neighbors verify that a borrower's Debt-Service Coverage Ratio satisfies \text{DSCR}_{i,t} \ge 1.25 and that the supplier is KYB-whitelisted in zero knowledge—without ever seeing the borrower's raw daily profit margins or family medical remittances[cite: 1, 2].

4. Direct-to-Supplier Programmable Escrow Wallet (Topic F +5 Bonus): Uses ERC-4337 Account Abstraction (seedless, gasless bounded session keys) and ‘KendraEscrow.sol’ on Base Sepolia & FISCO BCOS to lock peer social collateral and pay 98.5% of loan principal directly to KYB-whitelisted wholesalers upon 4-of-5 EIP-712 co-signing—eliminating loan diversion 100% and bypassing Bangladesh's $119/day consumer cash-out cap[cite: 1, 2].

5. Four-Tier Zero-Trust Security & "Zero LLM Math" Firewall: Isolates multimodal AI (Whisper / Gemini 2.5 Flash) inside an untrusted Tier-0 perception sandbox with regex/perplexity OCR prompt-injection stripping, while executing 100% of financial math, Spatio-Temporal & Wash-Trading Entropy (H_{st} \ge 0.65), and x402 / Intercepta counterparty screening in deterministic Python (‘risk_engine.py’)[cite: 1, 2].

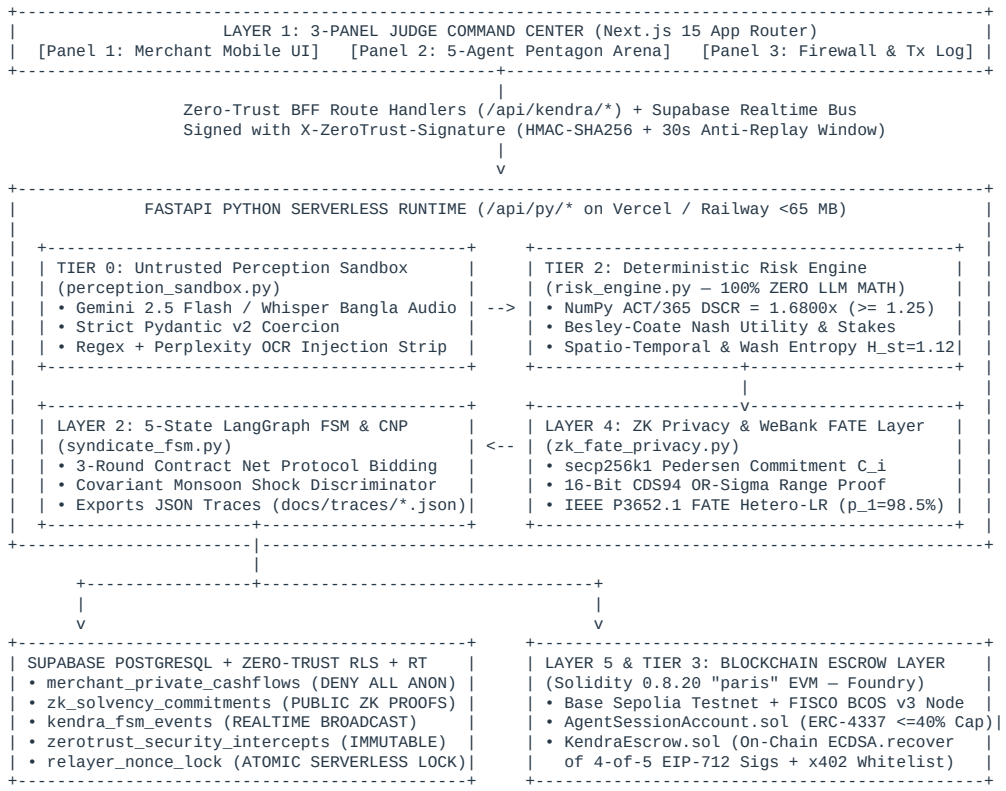
6. Yield-Bearing Social Collateral (‘ySCT’) & Autonomous Monsoon Flash Bailouts: Pays peer guarantors a 1.00% APR micro-guarantee yield and automatically executes an on-chain Implicit Joint-Liability Flash Bailout (‘triggerBesleyCoateBailout()’) + 14-day grace rescheduling when covariant weather shocks (e.g., Dhaka monsoon floods) hit the market[cite: 1, 2].

2. Project Architecture & Repository Structure

2.1 Repository Directory Structure

```
kendrafi/
■■■ CONTEXT.md                      # master system spec, 5 self-critic rules & reconciled math
■■■ vercel.json                     # unified Next.js 15 + Python serverless routing (<65 MB bundle)
■■■ app/
■   ■■■ page.tsx                    # 3-Panel Split-Screen Judge Command Center master layout
■   ■■■ demo/                       # 1-click Shenzhen judge scenario presets (A, B, C, Bailout)
■   ■■■ api/
■       ■■■ kendra/                 # Zero-Trust BFF routes (execute, bailout, repay, state)
■       ■■■ py/                     # FastAPI Python runtime (HMAC-guarded Tier-0 & Tier-2)
■           ■■■ perception_sandbox/ # Tier-0 Bangla voice (Whisper) & receipt OCR (Gemini Flash)
■           ■■■ risk_engine/        # Tier-2 Zero-LLM-Math NumPy DSCR, Besley-Coate & entropy
■           ■■■ zk_fate_privacy/    # Layer-4 secp256k1 Pedersen, 16-bit CDS94 range proof & FATE
■           ■■■ syndicate_fsm/      # Layer-2 5-state LangGraph FSM & 3-round CNP negotiation
■           ■■■ chain_executor/     # Layer-5 ERC-4337 session-key EIP-712 signer & Web3 relayer
■■■ components/
■   ■■■ header/                     # JudgeCommandHeader, Bilingual (EN/■■) HUD, 2.5x Zoom bar
■   ■■■ mobile/                     # Left Panel: Rahim/Sohel ERC-4337 phone, waveform, receipt scanner
■   ■■■ arena/                      # Center Panel: 5-Agent Pentagon SVG, ZK card, Besley-Coate table
■   ■■■ firewall/                   # Right Panel: 4-Tier security monitor, EIP-712 feed, Blockscout
■   ■■■ layout/  shared/  ui/       # Monte Carlo drawer, unit-economics modal, shadcn primitives
■■■ contexts/                       # Bilingual (EN/■■) language, 2.5x judge zoom, FSM state presence
■■■ hooks/                          # useKendraRealtime, useScenarioRunner, useFocusZoom, useBlockscout
■■■ lib/
■   ■■■ zerotrust/                  # HMAC-SHA256 request signer, 30s anti-replay, tier boundary guards
■   ■■■ crypto/                     # EIP-55 checksum helpers, EIP-712 typed data builders, Viem client
■   ■■■ audio/  ocr/                # Bangla voice waveform synthesizer, tampered receipt highlighter
■   ■■■ supabase/                   # server (service_role) & browser (anon Realtime) clients
■■■ contracts/
■   ■■■ foundry.toml                # solc 0.8.20 (evm_version = "paris") for Base Sepolia & FISCO BCOS
■   ■■■ src/                        # KendraEscrow.sol (ECDSA.recover), MockBDTc.sol, AgentSessionAccount.sol
■   ■■■ script/  test/              # DeployKendra.s.sol (15+ tx hash generator) & Foundry fuzz tests
■■■ supabase/                       # schema, seed, deny-all RLS, immutability triggers, migrations 001-003
■■■ public/                         # Bangla voice .mp3 samples, Meghna & Patuatuli receipt scans, icons
■■■ scripts/                        # generate-sandbox, monte-carlo-backtest, export-traces, verify-env
■■■ test-invoices/                  # sample valid Karwan Bazar rice invoices & Red-Team OCR injection photos
■■■ docs/                           # Technical Docs, Security Assessment, Deck, Video Script, JSON traces
■■■ types/index.ts                  # shared TypeScript types (FSMState, ZKCommitmentProof, PeerAgentVote)
```

2.2 High-Level System Topology



2.3 The 5 Modular Architectural Layers

Layer	Core Modules & Files	Production Tech Stack	Primary Responsibility
Layer 1: Judge Command Center UI & BFF	‘app/page.tsx’, ‘components/mobile/’, ‘components/arena/’, ‘components/firewall/’, ‘app/api/kendra/’[cite: 1, 2, 3]	Next.js 15 (App Router), React 19, TypeScript, Tailwind CSS, Framer Motion, Viem, Supabase JS[cite: 1, 2]	3-Panel Split-Screen UI, Bilingual (‘EN + ■■’) HUD badges, 2.5x Judge Camera Zoom, and HMAC-signed BFF API routing[cite: 1, 2].
Layer 2: Multi-Agent FSM & CNP Orchestrator	‘app/api/py/syndicate_fsm/’, ‘app/api/py/perception_sandbox/’[cite: 1, 2, 3]	Python 3.11+, FastAPI, LangGraph, Gemini 2.5 Flash / Whisper API, Pydantic v2[cite: 1, 2]	Enforces the 5-state FSM (‘IDLE_MONITORING’ \rightarrow ‘ESCROW_COSIGN_AND_SETTLE’), Tier-0 OCR firewall, and 3-round Contract Net Protocol negotiation[cite: 1, 2].
Layer 3: Deterministic Risk & Anti-Collusion Engine	‘app/api/py/risk_engine/’, ‘scripts/generate-sandbox.py’[cite: 1, 2, 3]	Pure NumPy (Zero LLM Math), Custom Regex/Perplexity Sanitizer, x402 Counterparty Filter[cite: 1, 2]	Computes volatility-haircut \text{DSCR}_{i,t}, Besley-Coate Nash utility E[U_j(\alpha_j^)], optimal stakes \alpha_j^, and Spatio-Temporal/Wash-Trading Entropy H_{st}[cite: 1, 2].
Layer 4: ZK Privacy, FATE & Supabase RLS Vault	‘app/api/py/zk_fate_privacy/’, ‘supabase/migrations/’[cite: 1, 2, 3]	‘ecdsa’ (‘secp256k1’), ‘pycryptodome’, Supabase PostgreSQL + Row-Level Security (RLS)[cite: 1, 2]	Isolates raw MFS cash flows behind deny-all RLS, generates ‘secp256k1’ Pedersen Commitments + 16-bit CDS94 range proofs, and computes FATE Hetero-LR survival scores (p_i)[cite: 1, 2].
Layer 5: Programmable On-Chain Escrow & AA	‘contracts/src/KendraEscrow.sol’, ‘MockBDTc.sol’, ‘AgentSessionAccount.sol’, ‘app/api/py/chain_executor/’[cite: 1, 2, 3]	Solidity ‘0.8.20’ (‘paris’ EVM), Foundry, OpenZeppelin, ERC-4337 Session Keys, Web3.py, Base Sepolia & FISCO BCOS[cite: 1, 2]	Verifies 4-of-5 EIP-712 signatures on-chain via ‘ECDSA.recover’, enforces ‘isKYBSupplier’, pays 98.5% principal directly to wholesaler, sweeps repayments, and runs flash bailouts[cite: 1, 2].

2.4 The 7-Gate Deterministic Verification Pipeline

Data never flows through unstructured agent loops; every request passes through seven strict verification gates[cite: 1, 2]:

1. Gate 1 (Perception Boundary): ‘MerchantMobilePanel’ \rightarrow ‘Tier0PerceptionSandbox’. Raw Bangla audio and receipt image are parsed into ‘ExtractedInvoiceIntent’ JSON with zero tool/wallet access[cite: 1, 2].

2. Gate 2 (Firewall & x402 Counterparty Screen): ‘Tier0PerceptionSandbox’ \rightarrow ‘DeterministicRiskEngine’. Parallel regex/perplexity sanitizer and x402 ‘isKYBSupplier[supplier]’ check run; violations halt immediately in

‘BLOCKED_ADVERSARIAL’ and log to Supabase ‘zerotrust_security_intercepts’ + on-chain ‘recordAdversarialIntercept()’[cite: 1, 2].

3. Gate 3 (ZK Range Proof & FATE Scoring): ‘DeterministicRiskEngine’ \rightarrow ‘Secp256k1PedersenRangeProver’. Filters circular wash transfers, verifies \text{DSCR}_{i,t} = 1.6800\text{x} \ge 1.25 and H_{st} = 1.12 \ge 0.65, computes FATE Hetero-LR p_1 = 98.5\%, and writes the 16-bit CDS94 range proof to Supabase ‘zk_solvency_commitments’[cite: 1, 2].

4. Gate 4 (3-Round P2P Syndicate CNP Arena): ‘Secp256k1PedersenRangeProver’ \rightarrow ‘KendraSyndicateFSM’. Peer agents ‘AGT-02..05’ verify the 16 bit-proofs in zero knowledge and execute a 3-round Contract Net Protocol negotiation (‘AGT-04’ caps at 40%, ‘AGT-05’ abstains, ‘AGT-02’ & ‘AGT-03’ step up to 35% and 25% at a 1.00% APR clearing yield)[cite: 1, 2].

5. Gate 5 (Bounded Session-Key EIP-712 Co-Signing): ‘KendraSyndicateFSM’ \rightarrow ‘ZeroTrustSessionKeySigner’. Borrower ‘AGT-01’ + Guarantors ‘AGT-04’, ‘AGT-02’, ‘AGT-03’ (4-of-5 quorum) generate EIP-712 typed signatures bounded by \alpha_j \le 0.40[cite: 1, 2].

6. Gate 6 (On-Chain Quorum Recovery & Direct B2B Settlement): ‘KendraChainRelayer’ \rightarrow ‘KendraEscrow.sol’. Contract verifies all 4 EIP-712 signatures on-chain via ‘ECDSA.recover’, locks ‘45,462.33 BDTc’ in peer social collateral, deducts the ‘1.50%’ wholesaler MDR (‘675.00 BDTc’), and transfers ‘44,325.00 BDTc’ (‘98.5%’) directly to Meghna Group (‘0x71C...A91’)[cite: 1, 2].

7. Gate 7 (Post-Loan Telemetry, Repayment Sweep & Flash Bailout): ‘KendraEscrow.sol’ \rightarrow ‘Supabase Realtime’ \rightarrow ‘SecurityEscrowPanel’. Streams live Blockscout receipt, schedules 1/15th-cost Bangla voice-bot daily MFS sweeps (‘repayFacility()’), and stands by for covariant monsoon shock recovery (‘triggerBesleyCoateBailout()’)[cite: 1, 2].

2.5 Four-Tier Zero-Trust Security & Supabase RLS Matrix

To satisfy the mandatory Security Self-Assessment Report[cite: 1, 2], KendraFi enforces four cryptographic permission tiers:

Permission Tier	Component / Actor	Allowed Operations	Cryptographic & Policy Boundary	Known Risk & Mitigation
Tier 0: Untrusted Perception	Multimodal LLM (Whisper / Gemini 2.5 Flash)[cite: 1, 2]	Parse Bangla voice & receipt images into raw JSON fields[cite: 1, 2].	Zero network, DB write, or wallet access; strict Pydantic v2 regex/perplexity stripping[cite: 1, 2].	Risk: Indirect OCR prompt injection[cite: 1, 2]. Mitigation: Parallel regex sanitizer + x402 registry lookup[cite: 1, 2].
Tier 1: Local Peer Agent	5 Merchant Syndicate Agents (‘AGT-01..05’)[cite: 1, 2]	Verify public ZK proofs; broadcast EIP-712 staking votes[cite: 1, 2].	ERC-4337 Session Key bounded (\alpha_j \le 0.40); Supabase RLS blocks access to ‘merchant_private_cashflows’[cite: 1, 2].	Risk: Compromised single agent node[cite: 1, 2]. Mitigation: 4-of-5 EIP-712 multisig verified on-chain via ‘ECDSA.recover’[cite: 1, 2].
Tier 2: Deterministic Risk Engine	Python Quantitative Gatekeeper (‘risk_engine.py’)[cite: 1, 2]	Compute \text{DSCR}_{i,t}, Besley-Coate utility E[U_j], and H_{st} entropy[cite: 1, 2].	Protected by ‘X-ZeroTrust-Signature’ (HMAC-SHA256 + 30s anti-replay); 100% Zero LLM Math[cite: 1, 2].	Risk: Wash-trading or Sybil ring[cite: 1, 2]. Mitigation: Circular A \to B \to A transfer discounting + H_{st} \ge 0.65 gate[cite: 1, 2].
Tier 3: Smart Contract Escrow	‘KendraEscrow.sol’ (Base Sepolia & FISCO BCOS)[cite: 1, 2]	Verify 4 EIP-712 signatures; lock stakes; pay ‘isKYBSupplier’ directly[cite: 1, 2].	Immutable Solidity modifiers (‘ECDSA.recover’, ‘isKYBSupplier[supplier] == true’, ‘ReentrancyGuard’)[cite: 1, 2].	Risk: Reentrancy or unauthorized payout[cite: 1, 2]. Mitigation: On-chain quorum check + direct pull/whitelist settlement[cite: 1, 2].

3. Reconciled Mathematical & Cryptographic Specification

3.1 Cash-Flow Adjusted Debt-Service Coverage Ratio (\text{DSCR}_{i,t})

After discounting circular A \to B \to A wash transfers to zero, the engine evaluates merchant i’s 90-day MFS inflows \text{CF}_{\text{in},i}(t) and restocking outflows \text{CF}_{\text{out},i}(t) over tenor \Delta\tau = 30 days, applying a 95% one-tailed downside volatility haircut (z_{0.95} = 1.645) and family subsistence floor C_{\text{sub},i}[cite: 1, 2]:

\text{CF}_{\text{proj},i} = (\hat{\mu}_{\text{net},i} - C_{\text{sub},i}) \cdot \Delta\tau - z_{0.95} \cdot \hat{\sigma}_{\text{net},i} \cdot \sqrt{\Delta\tau}

$$R_i = L_i \cdot \left(1 + r_{\text{APR}} \cdot \frac{\Delta\tau}{365}\right), \quad \text{DSCR}_{i,t} = \frac{\text{CF}_{\text{proj},i}}{R_i} \ge 1.25$$

3.2 Proportional Besley-Coate Staking (α_j^*) & Nash Expected Utility ($E[U_j]$)

Peers with coefficient of variation $CV_j = \sigma_{\text{in},j} / \mu_{\text{in},j} > 0.55$ or $p_j < 0.90$ abstain ('AGT-05' abstains)[cite: 1, 2]. Eligible peers are weighted by Ghatak (1999) assortative reliability $w_j = p_j \cdot \max(0, B_j)$, capped at $\alpha_j \le 0.40$, with any residual redistributed proportionally across uncapped active peers[cite: 1, 2]:

$$\alpha_j^{\text{raw}} = \frac{w_j}{\sum_{k \in \mathcal{N}_{-i}} w_k}, \quad \alpha_j^* = \min\left(0.40, \alpha_j^{\text{raw}} + \text{ProportionalResidual}_j\right)$$

Each peer guarantor j's expected utility $E[U_j(\alpha_j^)]$ incorporates the micro-guarantee yield $\phi_{\text{period}} = 0.010 \cdot \frac{\Delta\tau}{365}$, the discounted continuation value $\delta V_j(\alpha_j^)$ of saving 11.50\% APR (24.0\% \text{ MFI} - 12.5\% \text{ KendraFi}) on their progressive credit line (1.5 B_j), expected bailout loss, and volatility shadow cost[cite: 1, 2]:

$$\delta V_j(\alpha_j^) = 0.95 \cdot \left(\frac{0.240 - 0.125}{365} \cdot \Delta\tau\right) \cdot (1.5\, B_j) \cdot \alpha_j^$$

$$E[U_j(\alpha_j^)] = p_i \left(\phi_{\text{period}} \alpha_j^ L_i + \delta V_j(\alpha_j^)\right) - (1 - p_i) \alpha_j^ R_i - \lambda_{\text{vol}} CV_j^2 \alpha_j^* R_i \ge 0$$

3.3 Elliptic-Curve ('secp256k1') Pedersen Commitments & 16-Bit Disjunctive Range Proof

To prove $v - T \ge 0$ (where $v = \lfloor \text{CF}_{\text{proj},1} \rfloor = 76{,}376$\text{ BDT} and $T = \lfloor 1.25\, R_1 \rfloor = 56{,}828$\text{ BDT}, so $\Delta v = v - T = 19{,}548 \in [0, 2^{16}-1]$) without revealing v[cite: 1, 2]:

1. Compute borrower commitment $C_1 = v G + r_1 H$ and homomorphic excess commitment $C_{\text{excess}} = C_1 - T G = \Delta v G + r_1 H$ on 'secp256k1' (where H is derived via nothing-up-my-sleeve SHA-256 hash-to-curve).

2. Decompose $\Delta v = \sum_{k=0}^{15} b_k 2^k$ ($b_k \in \{0,1\}$) with $\sum_{k=0}^{15} 2^k r_{b_k} \equiv r_1 \pmod q$, generating 16 bit commitments $C_{b_k} = b_k G + r_{b_k} H$ and 16 Cramer-Damgård-Schoenmakers (CDS94) 1-out-of-2 OR-Sigma Proofs (e_0, z_0, e_1, z_1).

3. Peer agents verify all 16 bit-proofs and verify the homomorphic curve-point sum $\sum_{k=0}^{15} 2^k C_{b_k} == C_1 - T G$.

3.4 Spatio-Temporal & Wash-Trading Collusion Entropy (H_{st})

Combines Supplier Concentration Entropy (H_{sup}), MFS Arrival Timestamp Lag Entropy (H_{time}), Sender Diversity Graph Entropy (H_{sender}), and Maximum Pairwise Inflow Correlation ($\rho_{\max}$)[cite: 1, 2]:

$$H_{st} = 0.45 H_{\text{sup}} + 0.25 H_{\text{time}} + 0.30 H_{\text{sender}} - 0.50 \rho_{\max} \ge 0.65 \quad (\text{with } \rho_{\max} \le 0.85)$$

4. Key Notes & Extra Context for Teammates and AI Coding Agents

4.1 Single Source of Truth: Reconciled Numbers Cheat Sheet

CRITICAL FOR ALL AGENTS & TEAMMATES: Never hardcode conflicting numbers across the Python backend, Solidity tests, Next.js UI, or Pitch Deck. Always use these exact formula-derived values:

Metric / Parameter	Exact Reconciled Value	Mathematical Derivation / Note
Borrower Loan Principal (L_1)	'45,000.00 BDT' (~$400 USD)[cite: 1, 2]	Rahim Store ('AGT-01') 50 sacks Miniket rice restock for $\Delta\tau$ = 30 days[cite: 1, 2].
KendraFi Borrower Effective APR	'12.50% APR' (ACT/365)[cite: 1, 2]	'8.50%' (20/80 Capital) + '0.40%' (IT/Voice O&M) + '1.40%' (NPL Provision) + '2.20%' (1.20% Platform + 1.00% Peer Yield) + '0.00%' (Cash-Out Fee)[cite: 1, 2].

Metric / Parameter	Exact Reconciled Value	Mathematical Derivation / Note
Gross Repayment Obligation (R_1)	‘45,462.33 BDT’	45,000 \times (1 + 0.125 \times 30 / 365) = 45,462.33\text{ BDT} (Do NOT use ‘45,468’).
Daily Automated MFS Sweep	‘1,515.41 BDT / day’	45,462.33 / 30\text{ days}, nudged by 1/15th-cost Bangla voice bots[cite: 1, 2].
Wholesaler Flat MDR Discount	‘1.50% Flat’ (‘675.00 BDT’)[cite: 1, 2]	Deducted from principal upon instant B2B settlement; funds Paymaster & Bailout Reserve[cite: 1, 2].
Net Direct Payout to Supplier	‘44,325.00 BDTc’ (‘98.5%’)[cite: 1, 2]	Transferred directly from ‘KendraEscrow.sol’ to ‘0x71C...A91’ (‘0%’ loan diversion)[cite: 1, 2].
‘AGT-01’ 90-Day Cash-Flow Stats	\hat{\mu}_{\text{in}}=8,500, \hat{\mu}_{\text{out}}=3,450, C_{\text{sub}}=2,200, \hat{\sigma}_{\text{net}}=1,012.35	Yields \text{CF}_{\text{proj},1} = 76,376.71\text{ BDT} and exact \text{DSCR}_{1,t} = \mathbf{1.6800\text{x}} \ge 1.25[cite: 1, 2].
‘AGT-04’ (Nasrin Pharmacy) Stake	‘40.00%’ (‘18,184.93 BDT’)[cite: 1, 2]	Capped by Tier-1 session key (\alpha_4^* = 0.4000); E[U_4] = \mathbf{+169.85\text{ BDT}}[cite: 1, 2].
‘AGT-02’ (Fatema Textiles) Stake	‘35.00%’ (‘15,911.82 BDT’)[cite: 1, 2]	Steps up in CNP Round 2 (\alpha_2^* = 0.3500); E[U_2] = \mathbf{+64.12\text{ BDT}}[cite: 1, 2].
‘AGT-03’ (Kamal Agro) Stake	‘25.00%’ (‘11,365.58 BDT’)[cite: 1, 2]	Steps up in CNP Round 2 (\alpha_3^* = 0.2500); E[U_3] = \mathbf{+51.40\text{ BDT}}[cite: 1, 2].
‘AGT-05’ (Sohel Electronics) Stake	‘0.00%’ (‘0.00 BDT’ — Abstained)[cite: 1, 2]	High volatility CV_5 = 0.634 > 0.55 \implies E[U_5] = \mathbf{-142.60\text{ BDT}} < 0[cite: 1, 2].
WeBank FATE & Entropy Scores	p_1 = \mathbf{98.5\%}, H_{st} = \mathbf{1.12} \ge 0.65[cite: 1, 2]	Empirical synthetic backtest benchmarked against WeBank's +12% AUC uplift[cite: 1, 2].

4.2 The 5 Karwan Bazar Sandbox Personas (‘dhaka_kendra_90d.json’)

All wallet addresses MUST be passed through ‘Web3.to_checksum_address()’ before use in ‘web3.py’ or ‘viem’:

• ‘AGT-01’ — Rahim Store (Retail Grocery & Rice): Active Borrower in Scenario A; p_1 = 0.985; Primary KYB Supplier: Meghna Group Wholesale (‘MEGHNA-01’, ‘0x71C0000000000000000000000000000000000A91’)[cite: 1, 2].

• ‘AGT-02’ — Fatema Textiles (Tailoring & Fabrics): Lead Peer Guarantor (35% stake); p_2 = 0.970, B_2 = 48,000\text{ BDT}; Primary KYB Supplier: Islampur Fabric Mills (‘ISLAMPUR-02’, ‘0x82D0000000000000000000000000000000B42’)[cite: 1, 2].

• ‘AGT-03’ — Kamal Agro (Vegetable & Spice Stall): Peer Guarantor (25% stake); p_3 = 0.960, B_3 = 34,642.85\text{ BDT}; Primary KYB Supplier: Shyamoli Cold Storage (‘SHYAMOLI-03’, ‘0x93E0000000000000000000000000000000C18’)[cite: 1, 2].

• ‘AGT-04’ — Nasrin Pharmacy (Medicine Dispensary): Anchor Guarantor (40% capped stake); p_4 = 0.992, B_4 = 85,000\text{ BDT}; Primary KYB Supplier: Square Pharma Depot (‘SQUARE-04’, ‘0xA4F0000000000000000000000000000000D77’)[cite: 1, 2].

• ‘AGT-05’ — Sohel Electronics (Mobile Repair & Parts): Liquidity-Stressed / Adversarial Node; abstains in Scenario A (CV = 0.634) and acts as the active phone persona in Scenario B submitting a poisoned receipt to unverified Patuatuli Parts Hub (‘PATUATULI-99’, ‘0xDead0000000000000000000000000000Beef’)[cite: 1, 2].

4.3 Eight Non-Negotiable Engineering Guardrails for AI Agents

1. Keep Serverless Bundle <65 MB (No ‘scipy’ in ‘api/requirements.txt’): Vercel limits serverless functions to 250 MB unzipped. Use pure ‘numpy’ for all runtime math in ‘app/api/py/risk_engine/’. Reserve ‘scikit-learn’ strictly for the local CLI backtester (‘scripts/monte-carlo-backtest.py’)[cite: 3].

2. FISCO BCOS & Base Sepolia Bytecode Compatibility (‘solc 0.8.20’ + ‘paris’): Solidity ‘0.8.24’ defaults to the ‘shanghai’ EVM target (‘PUSH0’ opcode), which fails on standard FISCO BCOS v3 nodes. Always keep ‘solc_version = "0.8.20"’ and ‘evm_version = "paris"’ in ‘contracts/foundry.toml’.

3. On-Chain EIP-712 Signature Recovery (‘ECDSA.recover’): Do not rely solely on off-chain Python signature checks. ‘KendraEscrow.sol’ must verify all 4 EIP-712 signatures (‘1 Borrower + 3 Guarantors’) on-chain inside ‘executeSyndicateEscrow()’ and enforce ‘guarantorStakes[i] <= (grossRepayment * 4000) / 10000’.

4. Parallel Security Reporting in Scenario B: When Scenario B (‘SCENARIO_B_REDTEAM’) runs, do not let a Pydantic exception halt execution before the x402 address check runs. Catch the regex prompt-injection match, evaluate the x402

KYB whitelist check in parallel, and return both ‘TIER0_PROMPT_INJECTION_DETECTED’ and ‘X402_UNVERIFIED_SUPPLIER_ADDRESS’ so the Right Panel flashes both red badges simultaneously[cite: 1, 2].

5. Accurate Privacy & Tx Mode Badging: Because raw 90-day cash flows are stored in Supabase behind deny-all RLS (‘merchant_private_cashflows’) for cloud demo execution, badge the UI privacy shield as ‘[ZERO-TRUST RLS ISOLATED — SERVER/EDGE ONLY]’. Likewise, badge live RPC transactions as ‘[LIVE RPC BROADCAST]’ and any offline fallback as ‘[REPLAY OF VERIFIED TESTNET TX]’.

6. Supabase Realtime Requires ‘SELECT’ RLS Policies: Enabling RLS on ‘kendra_fsm_events’ without a ‘FOR SELECT TO anon, authenticated USING (true)’ policy causes Supabase Realtime to drop all broadcast messages silently. Keep ‘merchant_private_cashflows’ locked with zero ‘anon’ policies, and keep ‘kendra_fsm_events’ and ‘zk_solvency_commitments’ readable by ‘anon’.

7. Persona-Synced Left Panel Phone UI: When a judge clicks Scenario A, the Left Panel phone header displays Rahim Store (‘AGT-01’)[cite: 1, 2]. When a judge clicks Scenario B, the phone header switches to Sohel Electronics (‘AGT-05’ — Adversarial Node) so the UI persona and backend ‘borrower_agent_id’ are 100% consistent[cite: 1, 2].

8. Bilingual (‘EN + ■■’) Judge HUD: Keep Simplified Chinese technical badges visible alongside English headers (Weiyedai-Global ■■■■■■, FATE Hetero-LR ■■■■■■■■■■, 20/80 Joint Lending 20/80■■■■, Zero LLM Math ■■■■■■■■, ERC-4337 Escrow ■■■■■■) to maximize impact for Shenzhen judges[cite: 1, 2].

4.4 Quick Q&A Defense Notes for Judges

• If Tsinghua PBCSF Judges Ask About Grameen II Individual Liability vs. Domino Defaults: Explain that traditional binary (100%) joint liability causes strategic domino defaults when multiple members suffer a shock simultaneously (Besley & Coate 1995)[cite: 1, 2]. KendraFi solves this via Bounded Fractional Staking (\alpha_j \le 0.40) and Covariant Shock Discrimination: if \ge 4 of 5 Karwan Bazar merchants suffer a simultaneous >40\% MFS inflow drop during a monsoon flood, the engine classifies the shock as exogenous and executes a reserve-backed Flash Bailout + 14-day grace extension without slashing peer reputation[cite: 1, 2]. Peer slashing only triggers under idiosyncratic moral hazard[cite: 1, 2].

• If WeBank Architects Ask About Circular MFS Wash-Trading: Explain our two-layer defense: (1) ‘filter_circular_wash_transfers()’ + Sender Diversity Entropy (H_{\text{sender}}) inside ‘risk_engine.py’ discounts A \to B \to A wash loops to zero[cite: 1, 2]; and (2) ‘KendraEscrow.sol’ never disburses cash to the borrower—it settles ‘98.5%’ directly to KYB-whitelisted corporate wholesalers (Meghna Group, Square Pharma) who pay a ‘1.50%’ flat MDR fee, making cash extraction via wash-trading economically impossible without corporate supplier collusion (which is blocked by H_{\text{sup}})[cite: 1, 2].